

# Oladayo Robert Oyedeji

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## EDUCATION

### Columbia College

*Bachelor of Science in Computer Science, Mathematics, CGPA: 3.93/4.00*

Columbia, MO

*Jan 2023 – Dec 2026*

## PROFESSIONAL EXPERIENCE

### Undergraduate Research Assistant(NSF funded)

*University of Missouri*

May 2025 – July 2025

*Columbia, MO*

- Developed a real-time physiological signal-processing pipeline for wearable devices.
- Implemented filtering, feature extraction, and validation of heart-rate and respiratory metrics.
- Applied LLM methods to multimodal physiological emotion classification.

### Work Study Computer Science Assistant

*Columbia College*

Sep. 2024 – present

*Columbia, MO*

- Provisioned and maintained Fedora Linux VMs for systems programming coursework, configuring GCC toolchains and GDB debugging environments for 20+ students
- Led CS Club technical workshops on algorithms and software engineering, growing event attendance and fostering peer-to-peer learning.

## PROJECTS

### Flight-Telemetry-System 🐛 | *FreeRTOS, Embedded C, Git*

December 2025 – January 2026

- Architected a multithreaded satellite telemetry system with 5 concurrent FreeRTOS tasks (Telemetry, Sensor, Command, Environment, Watchdog) under priority-based preemptive scheduling.
- Designed fault-tolerant inter-task communication using FreeRTOS Queues and Mutexes, with a software Watchdog Timer enabling autonomous Safe Mode recovery.

### USB HID Gyroscope Controller | *Bare-metal C, STM32F411, TinyUSB, I2C, UART*

June 2026 – July 2026

- Implemented USB HID device firmware from scratch on STM32F411, handling descriptor enumeration, endpoint configuration and interrupt-driven data transfer
- Wrote bare-metal I2C driver to interface MPU-6050 IMU at register level, streaming gyroscope axes at 1kHz over USB HID reports

### RZIP compressor 🐛 | *C++, Compression, Git*

March 2026 – April 2026

- Implemented a from-scratch LZMA encoder featuring a hash-chain match finder, carry-safe range encoder with EWMA probability adaptation, and a 12-state Markov chain context model with 6000+ adaptive probability entries.
- Produced a valid LZMA bitstream achieving near-optimal entropy coding per Shannon's information theory across configurable dictionary sizes up to 4MB.

### MIPS/SPIM interpreter 🐛 | *C++, Computer Architecture, Git*

September 2025 – December 2025

- Engineered a full fetch-decode-execute pipeline for R, I, and J-type MIPS instructions using bitwise masking and manual ALU operations, with a 32-word register file and PC state machine
- Modeled realistic memory segments (text, data, stack) with alignment error detection, access violation handling, and deterministic state transitions

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Rust, MIPS Assembly, SQL (MySQL), JavaScript, STM32

**Build & Debug:** Linux (Fedora), FreeRTOS, CMake, Make, GDB, AddressSanitizer

**Frameworks:** OpenGL, SDL, PyTorch, Flask